

MiniSpeedCam r1.3 — PCB Design Review

Valar Systems — Hardware Engineering · Doc MSC-R1.3-DR-001 Rev A · 2026-06-22 · **CONFIDENTIAL**

Radar speed-camera motherboard (ESP32-S3 + STM32F301 + CDM324 24 GHz Doppler) · 4-layer, 50 × 50 mm Repo: `Valar-Systems/MiniSpeedCam @ version/r1.3` · Method: kicad-happy analyzer suite

Components	Stack	Routed	EMC risk	Errors / Warnings
84	4-layer (F / GND / PWR / B), 1.6 mm	100%	13 / 100	7 / 31

1. Scope & Method

The open-source `kicad-happy` analyzer suite was run against the r1.3 KiCad source (`version/r1.3/hardware/pcb/kicad/MiniSpeedCam-r1.3.{kicad_sch,kicad_pcb}`), and the results were triaged by hand against the raw netlist. Findings carry the analyzer rule IDs.

Analyzer	Status	Result
<code>analyze_schematic.py</code>	RAN	84 parts, 92 nets, 36 unique — 1 error / 5 warnings / 37 info
<code>analyze_pcb.py --full</code>	RAN	4-layer, 50×50 mm, fully routed (0 unrouted) — 6 errors / 26 warnings / 50 info
<code>cross_analysis.py</code>	RAN	5 plane-integrity findings (GND / 3V3 fragmentation)
<code>analyze_emc.py</code> (US market)	RAN	61 findings, EMC risk 13/100 (conservative)
<code>analyze_gerbbers.py</code>	SKIPPED	No Gerber/drill set exported yet — copper/aperture/drill DFM not yet checked (release gate, see §6)
Datasheet verification	PARTIAL	Front-end supply/ground pins confirmed against the netlist (§7); op-amp signal-pin order is a designer check
<code>analyze_thermal.py</code>	PARTIAL	Ran, but no power-budget input → junction temps not computed
SPICE	SKIPPED	No simulator on host

Layer stack: `F.Cu` / `In1.Cu = GND plane` / `In2.Cu = power plane` / `B.Cu` , 1.6 mm.

2. Executive Summary

The board is complete, fully routed, and mechanically tidy on a 50 × 50 mm 4-layer stack. The issue that matters most clusters around one theme: **the sensitive 24 GHz Doppler analog front-end is not isolated from the digital/RF noise of the ESP32-S3**. The radar amplifiers run from the digital `3V3` rail and reference the digital ground, and the isolated DC-DC that feeds the radar has its secondary ground bonded straight back to digital `GND` . This is the kind of coupling that forces firmware-side mitigation — the current firmware blanks the radar during WiFi bursts and runs WiFi at reduced TX power to keep readings usable. Fixing it in hardware is the highest-value content for this board.

There is also **one power-architecture item worth a deliberate decision** (§3.2): **three power sources — the Phoenix inlet J1, the barrel jack DC1, and USB-C VBUS (USB2) — land directly on the 5V net** with no ORing / reverse-blocking shown. If two can be plugged at once, that's a back-feed / contention path. The rest are standard pre-fab DFM and EMC hygiene items.

Top priority — Give the radar amplifier chain (U4 LMP7731, U12 MAX9814) a dedicated low-noise analog supply and a single-point analog ground, instead of the shared digital `3V3` / `GND` . This is the change most likely to remove the need for firmware radar-blanking and let WiFi TX power return to full range.

3. Priority Findings

3.1 Radar Front-End Shares the Digital Supply & Ground (HIGH)

```
RADAR SUPPLY [GOOD: dedicated, filtered diode rail]
5V → U5 B0505S (isolated DC-DC) → +Vo (pin4)
    → FB1 ferrite → C22 bulk → CDM324 V+

RADAR SIGNAL [uV signal amplified on the noisy digital rail]
CDM324 IF out → C11/R12 (AC couple)
    → U4 LMP7731 (gain) [U4.5 (V+) = 3V3]
    → U12 MAX9814 (AGC) [U12.2/5 (VDD) = 3V3]
    → ADC

GROUND: single GND net (no analog island); U5 secondary 0V tied to GND
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Observation	Evidence (netlist)	Severity
Doppler amplifiers on shared digital rail	U4.5 (V+) → 3V3 and U12.2/5 (VDD) → 3V3 — the same rail the ESP32-S3 / STM32 logic runs on. 3V3 is sourced from the U9 AP63203 switching regulator, so the amps see switching ripple as well as WiFi-TX transients. Supply pins confirmed (§7).	HIGH
Isolated converter ground bonded to digital ground	U5.1 (-Vin) → GND and U5.3 (0V) → GND ; no analog/isolated ground net exists. The B0505S isolation buys nothing for noise — the radar return shares digital ground.	HIGH
Diode rail done well	U5.4 (+Vo) → FB1 → C22 then on to the CDM324 — a separate, ferrite-filtered radar supply.	OK

Recommended changes

- Derive a clean analog rail for U4/U12 from the isolated B0505S +Vo via a small low-noise (high-PSRR) LDO — explicitly not the switcher-derived digital 3V3.
- Give the front-end a local analog ground island, tied to digital GND at a single star point near the STM32 ADC return.
- Decide deliberately whether the B0505S secondary 0V joins GND at one point (quiet) or stays isolated (true isolation). Today it's a common-ground bond, giving up most of the part's benefit.

Note: VOUT is the MAX9814 audio output to the ADC, not a supply — so RS-001 "VOUT has no source" is a false positive (§8).

3.2 Three Power Sources Merged on 5V (MEDIUM)

The 5V net is driven by three inlets in parallel — Phoenix J1.1, barrel jack DC1.4, and USB-C USB2.VBUS (pins 2/11) — plus U5.+Vin and the U9 switcher input all on the same node. The only series parts on the net are D1 and R1.

- If a user can have USB plugged and the barrel jack / Phoenix energized at the same time, current can back-feed from one source into another; at minimum the host USB port sees the external supply on its VBUS.
- Action: add ideal-diode ORing or Schottky reverse-blocking per source (or a deliberate "external or USB, never both" mechanical interlock / 0Ω select). Confirm D1's role — if it's the only reverse protection it should sit on each source, not just the rail.

4. Ground / Plane Integrity & EMC

Rule	Severity	Finding & action
PS-002	WARN	GND resolves into 5 copper islands, 2 signals crossing (GPIO43, GPIO44); 3V3 into 22 islands, 2 crossing. Verify In1.Cu is a continuous ground reference (no traces carved through it) and stitch the pours. 3V3 fragmentation matters because the analog amps reference 3V3.
GP-001	ERROR (x2)	Two signals cross a significant reference-plane gap (plus several partial). Reroute to avoid the void, or add a bridge cap if a split is intentional.
RP-001	ERROR (x2)	CAM.PCLK and CAM.XCLK change layers with no return-path stitching via (plus ~20 warning-level nets: camera data, UART, BOOT0). Add a ground via at each clock layer-transition.
IO-001	ERROR (x2)	No EMC filtering near the DC inlets / USB. Add TVS + pi/ferrite on the J1 Phoenix inlet and the DC1 barrel jack; add common-mode choke + ESD on the USB2 data pair.
CK-001 / CK-003	WARN	Camera XCLK / PCLK and SD_HOST_CLK on outer layers, routed near connectors J8/J10 — move inner-layer or add guard ground.
DC-003	WARN	~13 decoupling caps (C6/C7/C9/C11/C12/C17/C42...) far from their plane via — shorten cap→via path to cut PDN inductance.

EMC pre-compliance score 13/100 (conservative) — the analog-isolation and I/O-filtering gaps dominate.

5. DFM / Assembly

Rule	Severity	Finding & action
FD-001	ERROR	No fiducials either side (fine-pitch QFN present) — add ≥ 3 global fiducials.
PM-002	ERROR	C6 and FB1 sit 0.25 mm from board edge — move in ≥ 0.5 mm (crack risk on depanel). J1/J3 overhang is intentional edge-mount; J4 at 0.52 mm is borderline.
TE-001	WARN	0 test points / 91 nets. Add TPs on every rail, the radar analog output, and GND scope points.
DFM-001	WARN	0.1 mm / 0.116 mm trace+space forces JLC advanced tier. Relax non-camera nets to 0.127 mm to drop to standard if cost matters.
PM-001	WARN	Courtyard overlaps (U15/C2, U16/C26, U15/C56) — decoupling caps deliberately under their LDOs; confirm and waive.
TV-001	WARN	STM32 (U13) thermal pad has 5/9 recommended vias — top up to 9 (minor).
SS-001	ERROR	MPN coverage 3/36 unique parts (8.3%) in symbols — pre-fab sourcing blocker. Populate MPN fields from the LCSC numbers already in the symbols' <code>lcsc</code> property.

6. Schematic / Release Items

Rule	Severity	Finding & action
NT-001	WARN	STM32 <code>U13.PA2</code> and <code>U13.PA3</code> are single-pin (floating) bidirectional nets — add wires or no-connect flags so ERC is clean and the pins aren't left undefined.
RS-001 / pwr_flag	WARN	5V (and <code>1V2</code> / <code>2V8</code> / <code>3V3</code>) carry power_in pins but have no <code>PWR_FLAG</code> / power_out — ERC housekeeping; add a <code>PWR_FLAG</code> on the inlets.
GERBER	GATE	No Gerber/drill set committed. Export the fab package and re-run <code>analyze_gerbbers.py</code> before release — copper/aperture/drill DFM is otherwise unverified.

7. Datasheet Verification

Part	Pin checked	Result	Basis
U5 B0505S-1W	1 = -Vin, 2 = +Vin, 3 = -Vout(0V), 4 = +Vout	CONFIRMED	Standard B0505S 4-pin isolated DC-DC. <code>-Vin</code> and <code>0V</code> both on <code>GND</code> verified in netlist.
U12 MAX9814	5=VDD, 6=MICOUT, 8=MICIN, GND=4/7/9/10/11/15	CONFIRMED	MAX9814 14-pin TDFN. <code>VDD</code> → <code>3V3</code> verified.
U4 LMP7731	5 = V+	CONFIRMED	Pin 5 = V+ for SOT-23-5 single op-amp. <code>V+</code> → <code>3V3</code> verified.
U4 LMP7731	1-4 = OUT/+IN/-IN order	DESIGNER CHECK	Symbol uses standard SOT-23-5 order (1=OUT, 2=V-, 3=+IN, 4=-IN). Confirm visually against the LMP7731 connection diagram before fab. Not a flagged discrepancy.

8. Triaged as False-Positive / Expected (no action)

- **RS-001** on `VOUT` — `VOUT` is the MAX9814 audio output to the ADC, not a power rail.
- **CC-002** (~18x) "narrow signal" on camera / I²C nets — 0.1 mm is fine for low-current signals.
- **PM-001** courtyard overlaps (U15/C2, U16/C26, U15/C56) — decoupling caps deliberately under their LDOs.
- **DP-004** USB diff pair on outer layer — acceptable for this short full-speed run.

9. Review Gaps & Confidence

- **No Gerber check** (none exported) — copper / aperture / drill DFM not positively cleared.
- **No SPICE simulation** (no simulator on host): front-end gain/AGC/filter cutoffs not numerically verified (RC corners detected: R12/C11 $\approx$ 72 Hz, R12/C42 $\approx$ 1.25 MHz).
- **Thermal inconclusive:** no power-budget data, so junction temps not computed. Unlikely a blocker, but not positively cleared.

- **Datasheet sync partial:** front-end supply/ground pins verified; LMP7731 I/O pin order and the STM32/ESP pin maps not exhaustively checked against rendered figures.

Bottom line: the board is electrically functional and fully routed, but its radar receive path is coupled to the digital domain through both supply ([3V3](#)) and ground. Fixing that — a dedicated low-noise analog rail + single-point analog ground for U4/U12, a continuous ground reference, and basic I/O filtering — is the highest-value work, and directly targets the noise problem the firmware is currently working around. Before fabricating, also close the 3-source [5V](#) merge (\$3.2), export a Gerber set, and clear the standing DFM gates (fiducials, edge clearance, test points, MPN coverage).

Generated with the kicad-happy analyzer suite. Rule IDs reference kicad-happy detectors. Findings are engineering guidance for human review, not a substitute for DRC/ERC sign-off or full datasheet verification.